Summer Examinations 2019/2020

Course Instance Codes	2BDS1, 3BA1, 3BS9, 3BPT1, 4BS2, 4BME1, 1CSD1, 1CSD2
Exams	Second Arts with Data Science, Third Arts, Third Science, Third Bachelor of Science (Physics), Fourth Science, Fourth Mathematics and Education, MSc Data Analytics
Module Codes	**ST312**
Module	Applied Statistics II
External Examiner	Professor S. Wilson
Internal Examiner(s)	Dr. A. Simpkin Prof. J. Newell
Instructions:	**Answer ALL FIVE questions** **This is an OPEN BOOK exam** **Issues can be emailed to andrew.simpkin@nuigalway.ie**
Integrity:	**In submitting this work you confirm that it is entirely your own. You acknowledge that you may be invited to online interview if there is any concern in relation to the integrity of your exam.**
Duration	2 Hours
No. of Pages	15 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release in Exam Venue	Yes ☑ No ☐

Requirements:

Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper.

Question 1: [Total marks 15]

Does reading a book in bed make a difference to sleep in comparison to not reading a book in bed?

In a recent study, 90 people were recruited on a convenience basis from a local University. The participants were assigned to either read a book for 30 minutes before sleep or not to read a book before bed. All participants wore a device which objectively measured hours slept. Data were recorded at baseline and again 4 weeks later.

a) Is this an observational or experimental study? Explain your answer. (2 marks)
b) Identify the response and explanatory variables, the treatments, and the experimental units. (4 marks)
c) Are the results generalisable from this sample to the population of all individuals? Why/why not? (2 marks)
d) How should the researchers assign the subjects to the groups? Why? Explain the advantages of your proposed approach. (3 marks)
e) Does this experiment give a fair comparison between the two treatments? If not, why not? (2 marks)
f) How could this study design be improved? (2 marks)

Question 2: [Total marks 15]

A fast food franchise is test marketing three new menu items. To find out if the items have the same popularity, six franchisee restaurants are randomly chosen for participation in the study. In accordance with the randomized block design, each restaurant will be test marketing all three new menu items. Furthermore, a restaurant will test market only one menu item per week, taking 3 weeks to test market all menu items. The testing order of the menu items for each restaurant is randomly assigned.

Suppose each row in the following table represents the sales figures of the 3 new menu items in a restaurant after a week of test marketing

Restaurant	Item1	Item2	Item3
1	31	27	24
2	31	28	31
3	45	29	46
4	21	18	48
5	42	36	46
6	32	17	40

a) List the factor of interest and any nuisance factors. (2 marks)
b) What are the experimental units? (1 mark)
c) Are there blocks in this design? If so, identify them. (2 marks)
d) Why did the fast food franchise conduct the experiment in this way? (2 marks)
e) Has their approach been justified by the data? (2 marks)
f) Describe how a randomization test could be used to see if the items differ in mean sales, taking account of the possible between restaurant variation. State the null and alternative hypotheses being tested and describe (briefly) how such a test would be implemented including how the associated p-value would be calculated. What would be the advantage of doing this rather than a standard ANOVA F-test? (No calculations are required.) (6 marks)

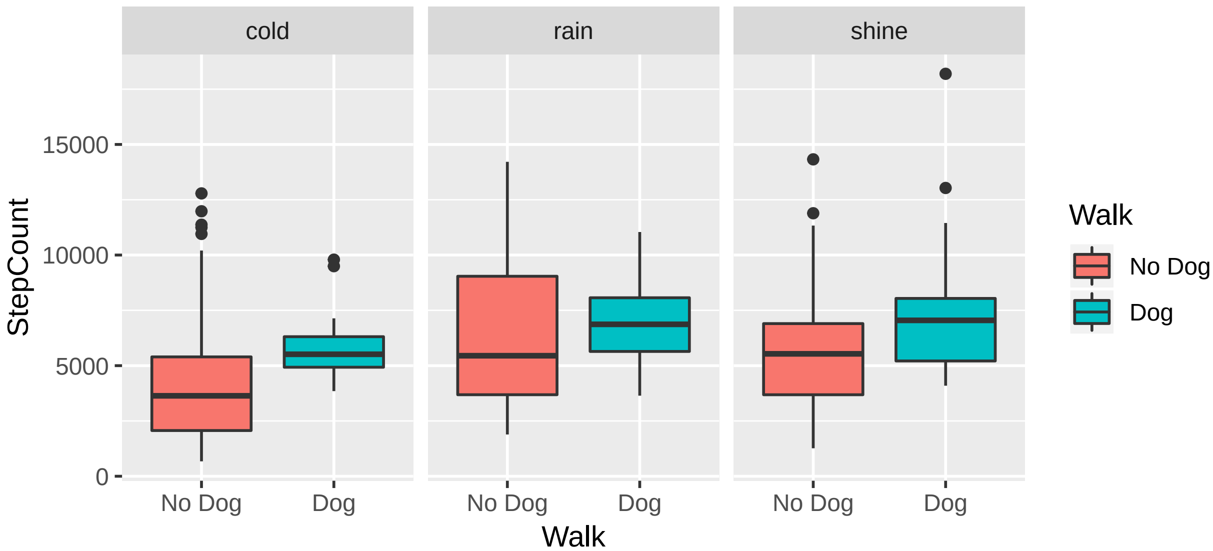
Question 3: [Total marks 30]

Is having a dog good for exercise? A researcher set out to investigate whether walking their dog had an effect on the number of steps they took. They took walks at different times with and without their dog (Dog v No Dog), each time recording the number of steps using a mobile app. The researcher also recorded the weather (cold, rain or shine) for each walk.

Descriptive statistics

```
# A tibble: 2 x 4
  Walk        n  mean    sd
  <fct>  <int> <dbl> <dbl>
1 No Dog   155 5391. 2942.
2 Dog       68 6837. 2488.
```

```
# A tibble: 3 x 4
  Weather     n  mean    sd
  <fct>   <int> <dbl> <dbl>
1 cold       69 4954. 2991.
2 rain       35 6631. 2877.
3 shine     119 6106. 2719.
```



a) An initial analysis using analysis of variance looked to see if there was any difference between walks with and without the dog. Use the output below to test the hypothesis that there was no difference between the mean number of steps with and without a dog. Clearly specify your null and alternative hypotheses, calculate an appropriate F-statistic and use the tables at the end of this paper to perform a test at the 5% level. (4 marks)

```
anova1 <- aov(StepCount ~ Walk, data = WalkTheDogs)
summary(anova1)
```

```
              Df     Sum Sq   Mean Sq F value    Pr(>F)
Walk           1   98924152 98924152       ?         ?
Residuals    221 1747586035   7907629
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

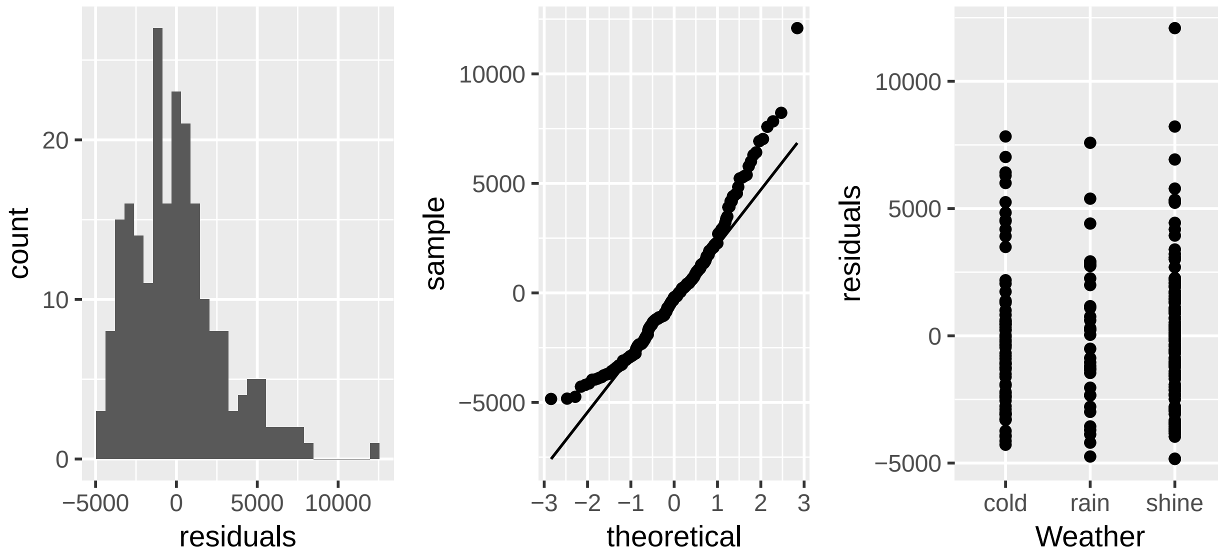
b) An alternative would have been to use a two-sample t-test; what would have been the value of the t-test statistic? (2 marks)

c) Next they looked to see if there was any difference due to weather. Using the output below, test the hypothesis that there are no differences between the weather types. Again, clearly specify your null and alternative hypotheses, perform an F-test at the 5% level, and state your conclusions. (4 marks)

```
anova2 <- aov(StepCount ~ Weather, data = WalkTheDogs)
summary(anova2)
```

```
              Df     Sum Sq   Mean Sq F value  Pr(>F)
Weather        2   84434667 42217333   5.271 0.00581 **
Residuals    220 1762075520   8009434
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d) Comment on the validity of the assumptions for this ANOVA based on the residual plots below. If you have reservations on the validity, suggest what you might do to improve matters. (3 marks)



e) Following on from the above ANOVA why does it make sense to perform a multiple comparisons procedure? (1 mark)

f) Use the following output to perform multiple comparisons. Group the different weather types accordingly and explain your decision using the relevant confidence intervals. (3 marks)

```
95 %-confidence intervals
 Method:  Difference of means assuming Normal distribution, allowing unequal variances


           estimate   lower  upper
rain-cold    1676.9   470.4 2883.4
shine-cold   1151.5   285.2 2017.8
shine-rain   -525.4 -1621.3  570.5
```

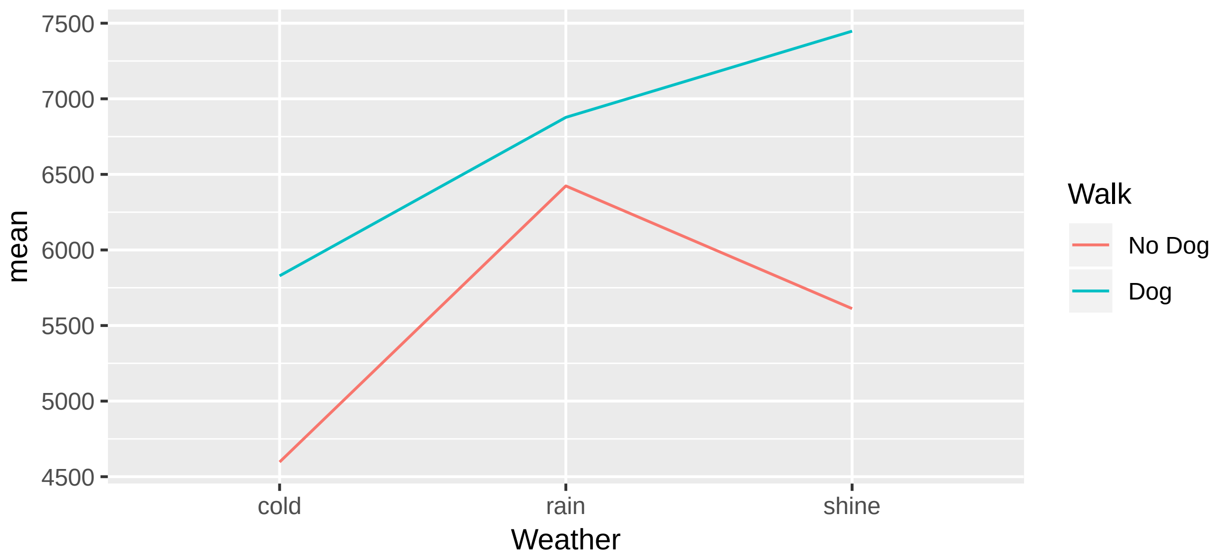
g) Define a suitable contrast to compare the population mean number of steps between rain and no rain (i.e. shine and cold combined). Specify the null and alternative hypotheses being tested, obtain an estimate for the contrast statistic, calculate its standard error and the degrees of freedom for the test. Perform the test at the 5% significance level. (6 marks)

h) To consider both factors, `Walk` (i.e. Dog v No Dog) and `Weather`, a two-way ANOVA was performed with the output given below. How does this test differ from the previous one way ANOVAs? What do you conclude? Relate your conclusions to your previous findings. (4 marks)

```r
anova3 <- aov(StepCount ~ Walk + Weather, data = WalkTheDogs)
summary(anova3)
```

```
              Df     Sum Sq  Mean Sq F value   Pr(>F)    
Walk           1   98924152 98924152  12.953 0.000395 ***
Weather        2   75066173 37533086   4.915 0.008168 ** 
Residuals    219 1672519862  7637077                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

i) Explain the interaction plot below. Does this suggest an interaction *might* be useful? Why? (1 mark)



j) The interaction model was fitted to the data, with output below. Interpret the result of the interaction model. (2 marks)

```r
anova4 <- aov(StepCount ~ Walk * Weather, data = WalkTheDogs)
summary(anova4)
```

```
              Df     Sum Sq  Mean Sq F value  Pr(>F)    
Walk           1   98924152 98924152  12.932 0.00040 ***
Weather        2   75066173 37533086   4.907 0.00824 ** 
Walk:Weather   2   12620439  6310219   0.825 0.43963    
Residuals    217 1659899424  7649306                    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4: [Total marks 15]

Brain tumours are thankfully rare, but researchers believe there may be a relationship between exposure to high voltage power lines in childhood and development of cancer in later life. They designed a case control study where they interviewed 12 patients with brain tumours and 12 healthy controls. The key questions was whether they had lived near power lines as a child. The following 2 x 2 table of results were found

	Power lines	No power lines	Total
Cancer	10	2	12
Healthy	3	9	12
Total	13	11	24

a) Compare cancer between those who lived by power lines and those who did not using the following measures and state what the relevant null and alternative hypotheses would be. You do not need to perform the test, simply calculate the test statistics:

(i) Equality of proportions;

(ii) `Relative Risk;`

(iii) Odds-ratio. (6 marks)

b) For small studies, such as this, standard large-sample results based on the normal distribution can be unreliable. Fisher's exact test is designed to deal with this problem and calculates a randomization p-value based on the observed table and table(s) with the same marginal totals that show stronger evidence of a treatment effect. List the table(s) that are more extreme than the above table, **calculate the p-value** and provide your conclusion for this study. (9 marks)

Question 5: [Total marks 25]

Researchers conducted a survey of 450 undergraduates in large introductory courses at either Mississippi State University or the University of Mississippi. There were close to 150 questions on the survey, but only three of these variables are included in this dataset. The primary interest for the researchers was factors relating to whether or not a student has ever overdrawn a checking account (0=no or 1=yes). The possible predictors of this are sex (0=male or 1=female) and number of days spent drinking alcohol in the past month.

```r
model1 <- glm(Overdrawn ~ DaysDrink,
              data = CreditRisk, family = binomial(link = "logit"))
summary(model1)
```

```
Call:
glm(formula = Overdrawn ~ DaysDrink, family = binomial(link = "logit"),
    data = CreditRisk)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.8944  -0.5414  -0.4531  -0.4303   2.2026

Coefficients:
            Estimate Std. Error z value             Pr(>|z|)
(Intercept) -2.33315    0.20871 -11.179 < 0.0000000000000002 ***
DaysDrink    0.05412    0.01691   3.201              0.00137 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 334.61  on 436  degrees of freedom
Residual deviance: 324.92  on 435  degrees of freedom
AIC: 328.92

Number of Fisher Scoring iterations: 4
```

a) Write down the fitted model in both the logit and probability forms of the logistic regression model. (3 marks)
b) Interpret the estimated slope (i.e. `Estimate`) for `DaysDrink` and the odds ratio for `DaysDrink` once you have calculated it. (4 marks)
c) Is there a statistically significant relationship between overdrawing and number of days drinking alcohol? State the null and alternative hypotheses, associated p-value, and your conclusions. (4 marks)
d) Use the fitted model to predict the probability of overdrawing for someone who drank alcohol on 7 days in the past month. (2 marks)

e) Calculate and interpret 95% confidence interval for the **odds ratio** of `DaysDrink`. (5 marks)

The previous analysis (`model1`) ignores possible differences between males and females. To explore this two more models were fitted as shown below.

```
model2 <- glm(Overdrawn ~ DaysDrink + Sex,
              data = CreditRisk, family = binomial(link = "logit"))
summary(model2)
```

```
Call:
glm(formula = Overdrawn ~ DaysDrink + Sex, family = binomial(link = "logit"),
    data = CreditRisk)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.1885  -0.5437  -0.4935  -0.2969   2.5076

Coefficients:
            Estimate Std. Error z value             Pr(>|z|)
(Intercept) -3.16896    0.35522  -8.921 < 0.0000000000000002 ***
DaysDrink    0.06901    0.01813   3.806             0.000141 ***
Sex          1.12487    0.33771   3.331             0.000866 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 334.61  on 436  degrees of freedom
Residual deviance: 312.43  on 434  degrees of freedom
AIC: 318.43

Number of Fisher Scoring iterations: 5
```

```r
model3 <- glm(Overdrawn ~ DaysDrink * Sex,
              data = CreditRisk, family = binomial(link = "logit"))
summary(model3)
```

```

Call:
glm(formula = Overdrawn ~ DaysDrink * Sex, family = binomial(link = "logit"),
    data = CreditRisk)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.3880  -0.4990  -0.4586  -0.3423   2.3944

Coefficients:
              Estimate Std. Error z value         Pr(>|z|)
(Intercept)   -2.84916    0.40329  -7.065 0.00000000000161 ***
DaysDrink      0.04118    0.02847   1.446            0.148
Sex            0.64971    0.47945   1.355            0.175
DaysDrink:Sex  0.04822    0.03732   1.292            0.196
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 334.61  on 436  degrees of freedom
Residual deviance: 310.73  on 433  degrees of freedom
AIC: 318.73

Number of Fisher Scoring iterations: 5
```

f) Use the output (specifically the deviance values) from the three model fits to fill in the analysis of deviance table below for the effects of `DaysDrink`, `Sex` and their interaction.

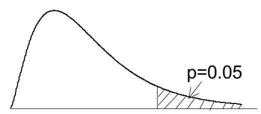
	Degrees of Freedom	Deviance	Drop in deviance	Critical value from $\chi^2_{0.05}$
Model1(days)				
Model2(days, sex, additive)				
Model3(days, sex, interaction)				

What do you conclude from this table? Justify your conclusion with a formal significance test. (5 marks)

g) Use the interaction model estimates to calculate the odds ratio for a one-day increase in number of drinking days for male and female students. (2 marks)

F distribution for p = 0.05

For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion p = 0.05 of the distribution falls. (Example, for df = 3,8 and F = 2.92 is exceeded by p = 0.05 of the population.

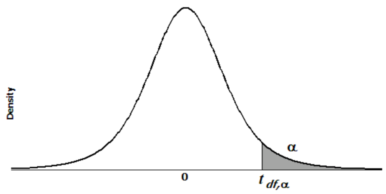


***p* = 0.05**

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

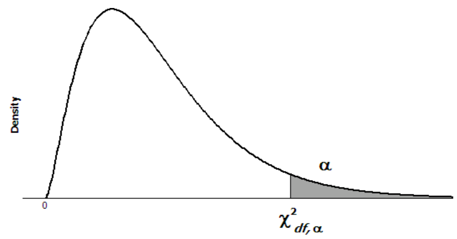
p.t.o.

Table: t distribution critical values



	Upper tail probability											
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.0025**	**0.001**	**0.0005**
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

Table: χ^2 distribution critical values



	Upper tail probability											
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.0025**	**0.001**	**0.0005**
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.52	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.69
80	88.13	90.41	93.11	96.58	101.88	106.63	108.07	112.33	116.32	120.10	124.84	128.26
100	109.14	111.67	114.66	118.50	124.34	129.56	131.14	135.81	140.17	144.29	149.45	153.17